USB-C

J1
USB-C (USB4105)
VBUS
CC1
CC2
D-
D-
D+
D+
SHIELD
GND
SBU1
SBU2
A4 USB_VBUS_IN
A5 USB_CC1
B5 USB_CC2
A7 USB_DN
B7
A6 USB_DP
B6
A8
B8
SH
A1
GND
GND
+5V
R14 5.1k
R13 5.1k
WALL_VBUS_IN
USB_DN
USB_DP
J3 USB-JST
MP
GND
D4 SS14
F1 1A fast (466)
USB_VBUS_IN
WALL_VBUS_IN
VBUS
SS14
D15
C22 100nF
D10 SMF5.0A
+5V
TPD2S017
CH1_OUT
GND
CH1_IN
CH2_OUT
VCC
CH2_IN
USB_ESP_DN
USB_DN
USB_ESP_DP
USB_DP
D5

ESP32-S3 MCU

+3V3
EN
BOOT
I2S_DOUT
I2S_WS
STATUS_LED
PTT_DRV
DOOR_DRV
MUTE_DRV
OC1_OUT
OC2_OUT
3V3
GND
TXD0
RXD0
USB_D-
USB_D+
PSRAM
USB_ESP_DN
USB_ESP_DP
I2C_SDA
I2C_SCL
I2S_MCLK
I2S_BCLK
I2S_DIN
U1
ESP32-S3-WROOM-1U-N16R8
C3 10uF
C6 100nF

POWER

U2
SGM2212-3.3
+5V
VIN
VOUT
GND
C2 10uF
C4 10uF
+3V3

STATUS LED

D6
STATUS
STATUS_LED
LED_A
R15 1k
R27 1k

BUS SWITCHES (PhotoMOS SSRs)

K1
GAQW212GS
PTT_DRV
R4 220
R7 10k
R24 220
K1_LED_A
K1_LED_B
P2
TALK_BRIDGE
TX_OUT
P3
R34 1.2kΩ
R35 1kΩ
HS_FILT
C25 22uF
C26 22uF
CF_RET
JP1 SolderJumper_2_Bridged
GND

TALK (K1, dual NO SSR)
ch1: P2 <-> Ra/Cf/Rb LPF -> TX_OUT
(the 2.2k talk handshake, gong-stripped);
ch2: TX_OUT <-> P3 (handshake + codec TX via C14);
cut JP1 = plain 2.2k strap (V4.1 fallback)

DOOR OPENER (K2, NO SSR)
bridges P2<->P3 directly (dead short),
emulating the WF26 OT button (S1)

K2
GAQY212GS
DOOR_DRV
R8 10k
R25 10M
D11 1N4148W
R5 220
K2_LED
R17 100k
DELAY_GATE
WD_GATE
C20 2.2uF
Q4 AO3400A
K2_CATH
Q3 AO3400A
C18 1uF
P3
P2
GND

K3
GAQY412EH
MUTE_DRV
R6 220
R9 10k
K3_LED
P4
CHIME_C1
GND

CHIME SUPPRESS (K3, NC SSR)
opens the C1 path (P4<->CHIME_C1);
mutes gong; line 4 + latch intact

K4
GAQY412EH
DOOR_DRV
R21 220
K4_LED
K5_COM
K5_COM_RLY
GND

Pull-down for DOOR_DRV is on K2

UNLATCH K5 (K4, NC SRR)

BELL SENSE

J2
WF26 bus
P1
P2
P3
P4
P5
D2 H24VND3BA
D3 H24VND3BA
D7 H24VND3BA
D12 H24VND3BA

HOUSE BELL / Tuerruf (P1->P4)

P4
D8 1N4148W
OC1
LTV-217 (PC817)
OC1_CATH
OC1_OUT
OC1_EMIT
R22 10k
R1 5.1k
GND

P5
D9 1N4148W
OC2
LTV-217 (PC817)
OC2_CATH
OC2_OUT
OC2_EMIT
R23 10k
R2 5.1k

APARTMENT BELL / Etagenruf (P1->P5)

RESET / BOOT

+3V3
R10 10k
EN
C5 1uF
GND
SW2
EN
R11 10k
BOOT
SW1
BOOT

AUDIO ES8311 + LINE XFMR

+3V3
R19 10k
C7 100nF
C8 100nF
FB1 600R@100MHz
AU_3V3
AVDD
C9 100nF
C10 10uF
+5V
C23 1uF
C24 1uF
U4
LP5907MFX-3.3
R18 10k
I2C_SDA
I2C_SCL
U3
ES8311
CCLK
MCLK
PVDD
DVDD
DGND
SCLK/DMIC_SCL
ASDOUT
LRCK
DSDIN
AGND
EP
CE
CDATA
MIC1P/DMIC_SDA
MIC1N
VMID
ADCVREF
DACVREF
OUTN
OUTP
AVDD
ES_CE
R20 10k
D13 BAT54S
D14 BAT54SW
ES_OUTP
ES_OUTN
ES_MICN
ES_MICP
ES_VMID
ES_ADCVREF
ES_OUTP_AC
ES_OUTN_AC
ES_MICN_AC
ES_MICP_AC
R26 22k
R16 22k
R31 22k
R30 22k
R32 3.3k
R31 3.3k
C11 10uF
C12 10uF
C13 1uF
C14 1uF
C15 1uF
C16 1uF
C17 1uF
TX_OUT
P1
P2

WF 26/G

SW3
Tueroeffner (door release)
SW4
Sprechen/Hoeren (talk)
MP1
MP2
K5_COM
K5_COM_RLY
D1 1N4148W
K5 G6K-2F-Y DC12
R29 2.2k
R29_BRIDGE
CHIME_C1
CHIME_MID
C19 47uF/50V
C21 47uF/50V
LS1
Speaker/Mic 16R

TEST POINTS

TP1
+3V3
TP2
TP3
TP4
TP5
TP6
TP7
TP8
GND
WD_GATE
AU_3V3
ES_OUTP
ES_MICP
ES_MICN
ES_VMID

H1 MountingHole
H2 MountingHole

Gunnar Beutner
Sheet: /
File: doorbell.kicad_sch
Title: Doorbell controller (Klingel V4)
Size: A3 | Date: 2026-07-02 | **Rev: V4.2**
KiCad E.D.A. 10.0.4 | Id: 1/1